Homework 2 — changed numbers (extra practice)

Module 2 · Homework · No date

The numbers on this sheet are changed from her sheet.

Name _____ Date _____

1. A 176-lb male received a single IV bolus dose of an antibacterial agent at a dose level of 15 mg/kg. The concentration of drug in the plasma was 75.5 mcg/mL and 36.8 mcg/mL at 2 and 8 hours, respectively, following administration of the dose. The antibacterial agent displays, linear, first-order, one compartment pharmacokinetics.

1a Module 2

What is the half-life of this agent in this patient?

1b Module 2

What is the expected initial plasma drug concentration of this dose of drug in this patient?

1c Module 2

What is the volume of distribution of this agent in this patient?

1d Module 2

How many hours following administration of the dose are required for 87.5% of the agent to be eliminated from the body?

1e Module 2

If the dose were administered at a level of 7.5 mg/kg, how many hours following administration of the dose are required for 87.5% of the drug to be eliminated from the body?

1f Module 2

What is the clearance of this agent in this patient?

2. The equation below represents the concentration of drug in the plasma following IV bolus administration of a 350 mg bolus dose. (Concentration is given in mcg/mL and time in hours)

$C_p = 8.72e^{-3.85t} + 5.40e^{-0.154t}$

2a Module 2

What is the elimination half-life of this agent?

2b Module 2

What is the initial concentration of drug in the plasma upon administration of the dose?

2c Module 2

What is the concentration of drug in the plasma 4 hours after administration of the dose?

2d Module 2

What is the apparent volume of distribution of the central compartment?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working.

1a

Worked by the drill

Answer: 5.79 hr

WHAT THE STEM GIVES

weight	176-lb	*not needed: the half-life comes from the two concentrations*
D_0	15 mg/kg	*not needed*
C_1	75.5 mcg/mL at 2 hours	the numerator inside the logarithm
C_2	36.8 mcg/mL at 8 hours	the denominator inside the logarithm

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{\ln(75.5/36.8)}{8\ \text{hr} - 2\ \text{hr}} = \frac{0.7186}{6\ \text{hr}} = 0.1198\ \text{hr}^{-1}$$

First order, so the rate constant is the log of the ratio of the two concentrations over the time between them.

ALGEBRA

$$t½ = \frac{0.693}{0.1198\ \text{hr}^{-1}} = 5.79\ \text{hr}$$

$t½ = \frac{0.693}{k}$, because $k\ t½ = \ln 2 = 0.693$ when the concentration falls to half. With $k = 0.1198\ \text{hr}^{-1}$ the reciprocal hours invert to hours: 5.79 hr per halving. Check: 2 to 8 hr is 1.04 half-lives, and the concentration fell from 75.5 to 36.8 mcg/mL, about half.

> **DOES THE NUMBER MAKE SENSE?**
> From 2 to 8 hr the level fell from 75.5 to 36.8 mcg/mL, almost exactly half, so the 6 hours between the samples is about one half-life: 5.79 hr, between 3 and 6.

Her homework wording with different numbers, so this is not the graded answer.

1b

Worked by the drill

Answer: 95.9 mcg/mL

WHAT THE STEM GIVES

weight	176-lb	*not needed: C_0 comes from extrapolating a measured level*
D_0	15 mg/kg	*not needed here; used with C_0 in part c*
C at 2 hours	75.5 mcg/mL	multiplied by e^{kt} to undo 2 hours of decline
C at 8 hours	36.8 mcg/mL	gave $k = 0.1198\ \text{hr}^{-1}$ in part a

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_0 = C\,e^{kt} = (75.5\ \text{mcg/mL})\,e^{(0.1198)(2)}$$

The one-compartment bolus line $C = C_0\,e^{-kt}$ rearranges to $C_0 = C\,e^{kt}$, where C_0 is the concentration the moment the dose is in. The first sample, 75.5 mcg/mL at 2 hr, has already declined for 2 hours at k = 0.1198 hr^{-1}; multiplying by e^{kt} undoes that decline.

ALGEBRA

$$C_0 = (75.5)(1.2707) = 95.9\ \text{mcg/mL}$$

kt = (0.1198 hr^{-1})(2 hr) = 0.2395, a pure number because the hours cancel, and $e^{0.2395}$ = 1.2707. (75.5 mcg/mL)(1.2707) = 95.9 mcg/mL: C_0 keeps the unit of the measured concentration and is higher than the 2 hr sample, as the level at time zero must be.

DOES THE NUMBER MAKE SENSE?

C_0 has to be above the 2-hour level of 75.5 mcg/mL, and 2 hours at k = 0.1198 hr^{-1} is about a third of a half-life, so the rise back is well under a doubling: 95.9 mcg/mL, between 75.5 and 2 × 75.5.

Her homework wording with different numbers, so this is not the graded answer.

1c

Worked by the drill

Answer: 12.51 L

WHAT THE STEM GIVES

weight	176-lb	converted: $\frac{176}{2.2} = 80.0$ kg
D_0	15 mg/kg	times 80.0 kg gives D_0 = 1200 mg, the numerator
C at 2 hours	75.5 mcg/mL	extrapolated in part b to C_0 = 95.9 mcg/mL, the denominator
C at 8 hours	36.8 mcg/mL	*not needed here: it served part a*

THE WORKING, ONE LINE AT A TIME

UNIT

$$176\ \frac{\text{lb}}{2.2}\ \text{lb/kg} = 80.0\ \text{kg};\ \text{dose} = (15\ \text{mg/kg})(80.0\ \text{kg}) = 1200\ \text{mg}$$

The dose is 15 mg for each kilogram of body weight, and the weight is in pounds; 2.2 lb is 1 kg, so $\frac{176\ \text{lb}}{2.2\ \text{lb/kg}} = 80.0$ kg, the pounds cancelling. (15 mg/kg)(80.0 kg) = 1200 mg: the kilograms cancel, leaving the whole dose D_0 in milligrams.

ALGEBRA

$$V_D = \frac{1200\ \text{mg}}{95.9\ \text{mg/L}} = 12.51\ \text{L}$$

Concentration is amount over volume, $C_0 = \frac{D_0}{V_D}$, so $V_D = \frac{D_0}{C_0}$: the volume of distribution is the apparent volume the 1200 mg dose occupies at 95.9 mcg/mL. 1 mcg/mL is 1 mg/L, so mg over mg/L cancels the milligrams and leaves litres, 12.51 L.

DOES THE NUMBER MAKE SENSE?

A 1200 mg dose giving 95.9 mg/L at time zero occupies $\frac{1200}{95.9}$, a little over 12 L: 12.51 L, above zero.

Her homework wording with different numbers, so this is not the graded answer.

1d

Worked by the drill

Answer: 17.36 hr

WHAT THE STEM GIVES

weight	176-lb	*not needed: the time to a fraction does not depend on the dose*
D_0	15 mg/kg	*not needed*
fraction eliminated	87.5%	leaves 12.5%, three half-lives
table	75.5 and 36.8 mcg/mL at 2 and 8 hours	gave t½ = 5.79 hr in part a

THE WORKING, ONE LINE AT A TIME

SETUP

87.5% eliminated leaves 12.5%: 100 → 50 → 25 → 12.5, three half-lives

87.5% eliminated means 100 − 87.5 = 12.5% of the dose remains. With first-order elimination each half-life t½ removes half of what is left: 100% to 50% after one, to 25% after two, to 12.5% after three, so the time to 12.5% remaining is three half-lives.

ALGEBRA

t = 3 × 5.786 hr = 17.36 hr

3 × 5.786 hr = 17.36 hr. The same result comes from $C = C_0\, e^{-kt}$ with C over C_0 = 0.125: $t = \frac{\ln 8}{k} = \frac{2.0794}{0.1198\ \text{hr}^{-1}}$ = 17.36 hr. The time to a stated fraction eliminated depends on k alone, not on the dose.

DOES THE NUMBER MAKE SENSE?

87.5% gone leaves 12.5%, which is 100 → 50 → 25 → 12.5, three halvings, so the time is three half-lives: 3 × 5.786 = 17.36 hr, more than the 6 hours between the two samples.

Her homework wording with different numbers, so this is not the graded answer.

1e

Worked by the drill

Answer: 17.36 hr

WHAT THE STEM GIVES

D_0	7.5 mg/kg	*not needed: a first-order time to a fraction does not*

		depend on the dose
fraction eliminated	87.5%	leaves 12.5%, three half-lives
table	75.5 and 36.8 mcg/mL at 2 and 8 hours	gave t½ = 5.79 hr in part a

THE WORKING, ONE LINE AT A TIME

SETUP

Halving the dose does not change k, so t = 3 × 5.786 hr = 17.36 hr

With first-order elimination the same fraction is lost per hour at any dose, so the time to lose 87.5% is unchanged.

> **DOES THE NUMBER MAKE SENSE?**
> Halving the dose halves every concentration but not the fraction lost per hour, so the time to lose 87.5% is still three half-lives: 17.36 hr, the same as part d.

Her homework wording with different numbers, so this is not the graded answer.

1f

Worked by the drill

Answer: 1.5 L/hr

WHAT THE STEM GIVES

weight	176-lb	with the dose gave V_D = 12.51 L in part c
D_0	15 mg/kg	with the weight gave the 1200 mg behind V_D
table	75.5 and 36.8 mcg/mL at 2 and 8 hours	gave k = 0.1198 hr^{-1} in part a

THE WORKING, ONE LINE AT A TIME

SETUP

$Cl = k \times V_D = (0.1198\ hr^{-1})(12.51\ L) = 1.50\ L/hr$

Clearance Cl is the volume of plasma cleared of drug per unit time; with first-order elimination the fraction k of the drug in the body leaves each hour, and that drug was spread through V_D, so $Cl = k \times V_D$. (0.1198 hr^{-1})(12.51 L) = 1.50 L/hr, reciprocal hours times litres.

> **DOES THE NUMBER MAKE SENSE?**
> The fraction k = 0.1198 of the 12.51 L leaves each hour, so about an eighth of the volume is cleared per hour: 1.50 L/hr, well below the 12.51 L volume.

Her homework wording with different numbers, so this is not the graded answer.

2a

Worked by the drill

Answer: 4.5 hr

WHAT THE STEM GIVES

D_0	350 mg bolus dose	*not needed: half-life does not depend on the dose*
A	8.72	*not needed: an intercept, not a rate*
α	3.85	*not needed: the larger exponent is distribution*
B	5.40	*not needed*
β	0.154	the denominator of $\frac{0.693}{\beta}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$t½\beta = \frac{0.693}{\beta} = \frac{0.693}{0.154\ \text{hr}^{-1}} = 4.50\ \text{hr}$$

The elimination phase is the smaller exponent, β; the larger, α, is distribution and is not the half-life asked for.

> **DOES THE NUMBER MAKE SENSE?**
> β = 0.154 hr^{-1} removes about 14% an hour, so halving takes between 4 and 5 hours: 4.50 hr. The α term, with a half-life well under an hour, is gone long before.

Her homework wording with different numbers, so this is not the graded answer.

2b

Worked by the drill

Answer: 14.12 mcg/mL

WHAT THE STEM GIVES

D_0	350 mg bolus dose	*not needed: the intercepts already carry the dose*
A	8.72	added to B
α	3.85	*not needed: at t = 0 both exponentials equal 1*
B	5.40	added to A
β	0.154	*not needed: at t = 0 both exponentials equal 1*

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_p^{\,0} = A + B = 8.72 + 5.40 = 14.12\ \text{mcg/mL}$$

$C_p = A\,e^{-\alpha t} + B\,e^{-\beta t}$ gives the concentration at any time; at t = 0 both exponents are zero and $e^0 = 1$, so each term equals its intercept and $C_p^{\,0} = A + B = 8.72 + 5.40 = 14.12$ mcg/mL, the concentration the moment the 350 mg bolus is in.

> **DOES THE NUMBER MAKE SENSE?**
> At time zero both exponentials are 1, so $C_p^{\,0}$ is the plain sum 8.72 + 5.40 = 14.12 mcg/mL, above either intercept alone.

Her homework wording with different numbers, so this is not the graded answer.

2c Worked by the drill

Answer: 2.92 mcg/mL

WHAT THE STEM GIVES

D_0	350 mg bolus dose	*not needed*
A	8.72	the coefficient of $e^{-\alpha t}$
α	3.85	the exponent $\alpha t = 3.85 \times 4 = 15.4$
B	5.40	the coefficient of $e^{-\beta t}$
β	0.154	the exponent $\beta t = 0.154 \times 4 = 0.616$
t	4 hours	t in both exponentials

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_p = 8.72e^{-(3.85)(4)} + 5.40e^{-(0.154)(4)} = 8.72e^{-15.4} + 5.40e^{-0.616}$$

The two-compartment equation $C_p = A\,e^{-\alpha t} + B\,e^{-\beta t}$ holds at any time, so both terms are evaluated at t = 4 hr. α = 3.85 hr^{-1} is the distribution rate constant and β = 0.154 hr^{-1} the elimination rate constant; each exponent is hr^{-1} times hours, a pure number: 15.4 and 0.616.

ALGEBRA

$$C_p = 0.00000 + 2.917 = 2.92 \text{ mcg/mL}$$

By 4 hours the distribution term is almost gone, so the elimination term carries nearly all the concentration.

> **DOES THE NUMBER MAKE SENSE?**
>
> By 4 hours the α term, $e^{-15.4}$, is zero to five decimals, so only the B term is left; βt = 0.616 is a bit under 0.693, so a bit over half of 5.40 remains: 2.92 mcg/mL, below 5.40.

Her homework wording with different numbers, so this is not the graded answer.

2d Worked by the drill

Answer: 24.79 L

WHAT THE STEM GIVES

D_0	350 mg bolus dose	the numerator
A	8.72	added to B for $C_p{}^0$
α	3.85	*not needed*
B	5.40	added to A for $C_p{}^0$ = 14.12 mg/L in part b, the denominator
β	0.154	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$$V_p = \frac{D_0}{A + B} = \frac{350\ \text{mg}}{14.12\ \text{mg/L}} = 24.79\ \text{L}$$

The central compartment is the volume the dose first distributes into, so $V_p = \frac{D_0}{C_p{}^0}$ with $C_p{}^0 = A + B = 14.12$ mcg/mL, the concentration at time zero. 1 mcg/mL equals 1 mg/L, so 350 mg over 14.12 mg/L cancels the milligrams and leaves litres, 24.79 L.

DOES THE NUMBER MAKE SENSE?

350 mg producing 14.12 mg/L at time zero fills $\frac{350}{14.12}$, about 25 L: 24.79 L, above zero.

Her homework wording with different numbers, so this is not the graded answer.